

Dilip Kumar

AI/ML ENGINEER · ASPIRING AI RESEARCHER & MODEL TRAINER

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SUMMARY

Second-year B.Tech student who ships AI systems end to end. I wrote an offline voice-typing app for Windows in Rust that runs Whisper on your own machine, and a multi-agent platform in Python with a React dashboard on top of it. What I actually care about is the training side: building the datasets, running the fine-tunes, and pre-training a small language model from scratch so I understand what everyone else is importing.

SKILLS

Languages	Python · Rust · TypeScript · C/C++ · SQL
ML / Deep learning	PyTorch · TensorFlow · scikit-learn · NumPy · Pandas · OpenCV
GenAI / LLMs	Transformers · Hugging Face · LangChain / LangGraph · RAG + vector DBs · QLoRA · Whisper (on-device)
Data & training	Dataset curation · Tokenizers · Eval harnesses · Synthetic data
Frameworks & cloud	React 19 · Tauri 2 · Docker · FastAPI · AWS · Linux · CI/CD
Security	Kali Linux · Burp Suite · OWASP Top 10 · TryHackMe

PROJECTS

Bolo · Offline voice typing for Windows

Rust, Tauri 2, whisper.cpp · **Shipped** · v0.10 · MIT

- Built an offline push-to-talk dictation app for Windows in Rust and Tauri 2, running whisper.cpp in-process so no audio ever leaves the machine.
- Cut word error rate from 13.2% to 9.6% on a 41-clip test set by adding a personal-vocabulary dictionary; 5.0% on clean English, ~1.9 s p50 to text on laptop CPU.
- Gated every release behind an evaluation harness, and shipped 10 public versions with a 3 MB installer.

github.com/Dilip-kumar-22/bolo

Typing Master · Touch-typing tutor, installable and offline

TypeScript, Preact, Vite, PWA · **Live** · v2.1 · MIT

- Shipped an installable, offline-first typing tutor: 41-chapter curriculum, adaptive drills weighted to the keys the user fails, colour-coded finger guide and a 3D keyboard heatmap.
- Kept all progress on-device with no account, added daily challenges and code-typing drills, and covered it with 137 passing tests.

dilip-kumar-22.github.io/typing-master-scorp/

S-CORP · An AI company that runs itself

Python, React 19, TypeScript, Docker · **Private beta** · v13

- Architected an autonomous multi-agent platform: a router agent dispatches intent to named specialists, and an autonomous engine generates, queues and executes work across an Inbox → Running → Review board.
- Wired 38 models across four providers behind one router, with 54 health-checked connectors, 18 scheduled jobs and Docker-sandboxed execution; runs ~4.2M tokens/day with live cost and latency metering.
- Built the React 19 command dashboard, and split the system into a three-repo monorepo (core / backend / frontend).

ORBIT · 3D scrollytelling starter, given away

Three.js, GLSL, OKLCH · **Live** · **Open source** · MIT

- Released an open-source 3D scrollytelling starter with HDR-safe post-processing (HalfFloat + 4× MSAA), an OKLCH palette shared between CSS and WebGL, and reduced-motion and no-WebGL fallbacks built in.

dilip-kumar-22.github.io/orbit/

FRIDAY · A terminal assistant that remembers

Python, Gemini, SQLite, Embeddings · **Private** · F-OS 1.0

- Built a terminal-native assistant on a three-tier memory architecture: a 1M-token working context, a self-compacting markdown journal, and a 412 MB store of 18,402 facts retrieved by hybrid vector + BM25 search.
- Routed specialist sub-agents by trigger rather than by prompt, and added a think mode that prints the reasoning before the answer so failures are debuggable.

RESEARCH & MODEL TRAINING

- Pre-training a custom ~200M-parameter language model from scratch: tokenizer, training loop and evaluation.
- Fine-tuning 70B-class open models with QLoRA on curated domain data.
- Dataset curation and eval-first pipeline design: deduplication, filtering, and writing the evaluation before the model.
- One arXiv paper a week, re-implemented in PyTorch.

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML)

Lovely Professional University, Punjab, India
2025 – 2029 (expected) · CGPA 9.00

CERTIFICATIONS

AWS Cloud Practitioner — in progress, 2026
Targeted next: AWS Solutions Architect Associate (2027), DeepLearning.AI Deep Learning Specialization (2027), AWS ML Specialty (2028).